Chenqi Meng

PhD student in Physics | The Chinese University of Hong Kong

mcqgravity@gmail.com | chenqitrg.github.io | Google Scholar | GitHub

RESEARCH INTERESTS

Generalized symmetries and classification of quantum phases; fusion categories and non-invertible SPT phases; tensor network renormalization and many-body simulation; algebraic formulations of conformal field theory and holographic duality.

EDUCATION

The Chinese University of Hong Kong | PhD in Physics, in progress

September 2021 - Present | Advisor: Zhengcheng Gu

University of Science and Technology of China | BS in Physics

September 2017 - May 2021 | GPA: 3.69/4.3

PUBLICATIONS AND PREPRINTS

Algebraic Tensor Network Renormalization and Holographic duality

Chenqi Meng, Tian Lan, Zhengcheng Gu.

Preprint (2026) | arXiv:2610.08787

A Practical Introduction to Tensor Network Renormalization with TNRKit.jl

Victor Vanthilt, Adwait Naravane, **Chenqi Meng**, Atsushi Ueda.

SciPost Physics Codebases 77 (2026) | DOI · arXiv:2604.06922

Non-invertible SPTs: an on-site realization of (1+1)d anomaly-free fusion category symmetry

Chenqi Meng, Xinping Yang, Tian Lan, Zhengcheng Gu.

Preprint (2024; revised 2025) | arXiv:2412.20546

RESEARCH SOFTWARE

TNRKit.jl | Collaborative Julia package for symmetry-aware tensor network renormalization, including TRG, HOTRG, and LoopTNR. Repository

BoundaryBulkKit.jl | Julia tools for Drinfeld centers, forgetful functors, and half-braidings using TensorKit.jl. Repository

SKILLS

Programming: Julia, Python, C, Mathematica.
Scientific computing: Tensor network algorithms, Loop-TNR, DMRG, many-body simulation.
Mathematics: Category theory, representation theory, conformal field theory, algebra.

OTHER PROJECTS

PeerPedia Core: A peer review CLI and library. Repository
Euclidean Reasoning Machine: An experiment with collaborating language-model agents for mathematical reasoning. Repository